LONGITUDINAL ANALYSIS OF PSYCHIATRIC MORBIDITY AND RESOURCE GAPS: A CASE STUDY OF KARNATAKA DISTRICTS (2018–2022)

1. Project Overview and Objective

This project creates a unified "District Health Fact Sheet" for Karnataka. By merging multiple distinct datasets, it transforms raw administrative records into an interactive analytical tool. The project tracks the prevalence of various mental health conditions across all districts of Karnataka over a 4-year period, providing a comprehensive view of the state's mental health burden versus its institutional capacity.

The primary goal is to move beyond simple "counting" and provide "context" through data integration.

- **Objective 1**: Trend Identification - To visualize the year-on-year growth or decline of specific mental health conditions (SMD, CMD, Substance Abuse) across Karnataka's districts from 2018 to 2022.
- **Objective 2**: Resource Gap Analysis - To calculate the Patient-to-Provider ratio by correlating district-level patient volumes with state-level mental health professional counts (Psychiatrists, Psychologists, and Nurses).
- **Objective 3**: Socio-Economic Correlation - To investigate if there is a statistical relationship between district literacy rates/alcohol consumption (from NFHS-5) and the reported cases of mental health disorders or alcohol abuse.
- **Objective 4**: Financial Efficiency Mapping - To assess the Budget per Patient metric by merging State Budget Approvals with the total patient load to see how mental health funding translates to the district level over time.
- **Objective 5**: Data Storytelling & Dashboarding - To build a Power BI dashboard that allows health administrators to filter by district and year, identifying "High-Burden Districts" that may require urgent resource intervention.

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2. Data Sources

- **Source Description and Timeline:** This project utilizes a multi-dimensional dataset consolidated from official government reports to analyze the mental health landscape in Karnataka. The data is integrated from the following primary sources taken from https://www.data.gov.in/: District-Level Patient Records (2018-2022); State Health Budgetary Approvals; Healthcare Infrastructure & Workforce; Socio-Demographic Indicators (NFHS-5).
- **Domain**: Healthcare [Psychiatry]

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3. Problem Statement

- **The Problem**: Despite the expansion of the National Mental Health Programme (NMHP), there is a significant information silo between patient incident rates at the district level and the allocation of financial and human resources at the state level.

1) Hidden Trends: We don't know if mental health issues like "Alcohol Abuse" or "Severe Disorders" are going up or down over the years because the data is stuck in 4 different yearly folders.
2) Missing Links: We don't know if the districts with the most patients actually have enough doctors and nurses to treat them.
3) Budget Confusion: It is unclear if the government is sending more money to the districts that are struggling the most or if the money is being spent elsewhere.

- **The Goal**: Review yearly patient growth trend; identify "High-Risk Districts" where patients are high but doctors are few; and help the government decide exactly where to send more funding and staff next year.

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4. Attribute (Column/Features) Details

Attribute Name	Data Type	Description
District	Text	Fundamental administrative unit, subdivision of State
State	Text	Geographical Location
Financial_Year	Text	Fiscal Year denoting April to March
Date	Date	Start Date of Each Year
Condition_Name	Text	Mental illness
Patient_Count	Integer	Total patients admitted
Pandemic_Status	Categorical	Time period based on severity
Severity	Categorical	Seriousness of the issue
Severity_Score	Integer	Quantitative metric of severity

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, Pivot Tables, Correlation Analysis, and ANOVA.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel/Power Query)

Tasks Performed:

- Converted multiple raw data obtained from https://www.data.gov.in/ into Fact and Dimension Tables and relationships were created.
- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, created conditional columns, created calculated fields, and applied conditional formatting.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
- **Analysis:** Performed Correlation Analysis and ANOVA to arrive at initial insights

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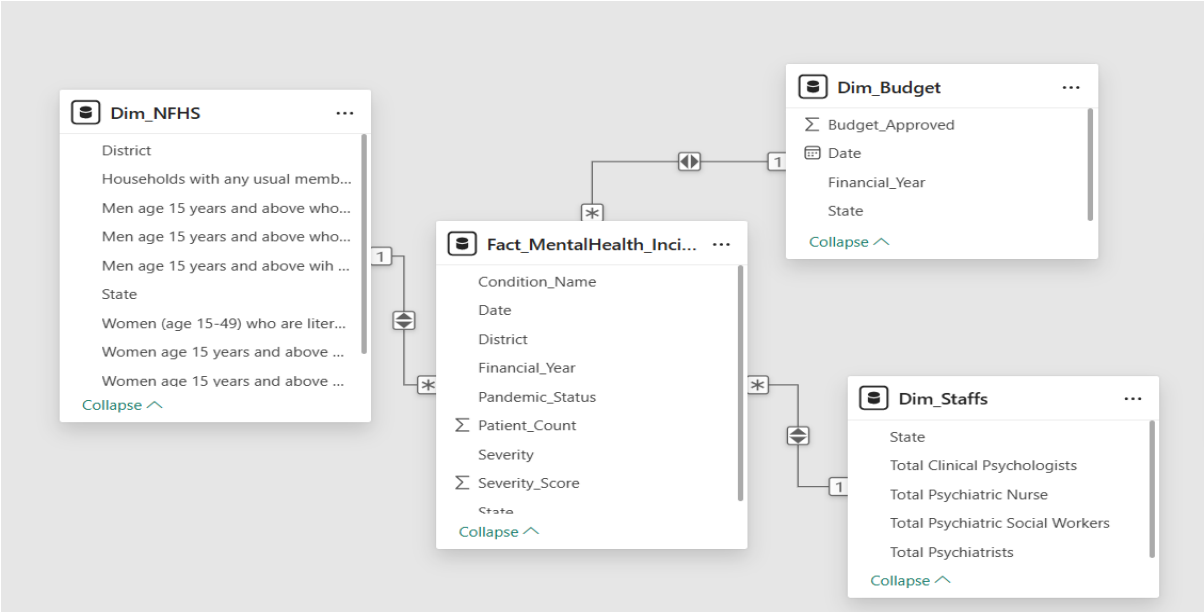
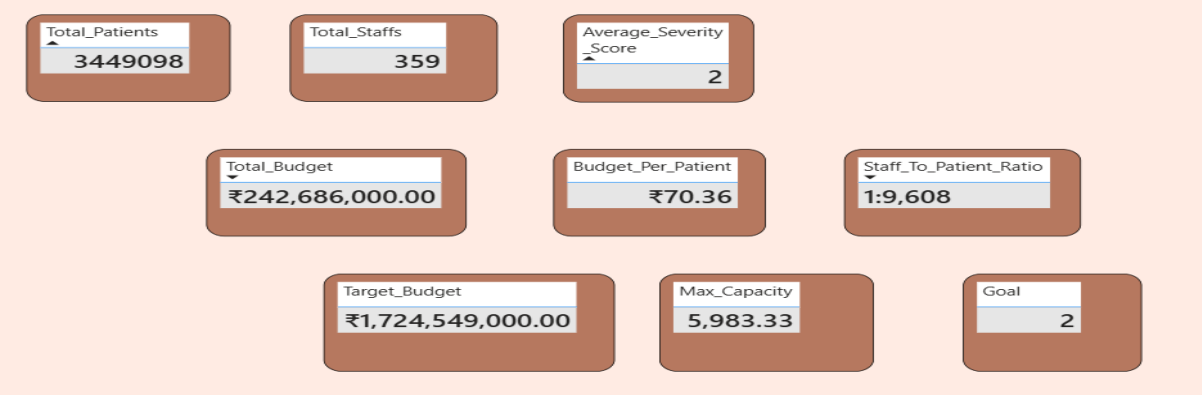
7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- **DAX Measures:** Implemented DAX formulas for key metrics, such as Total Patients, Total Budget, Average Severity Index, Total Staff, Staff to Patient Ratio, Budget per Patient, Max Capacity for Staff Calculation, Goal to check severity.



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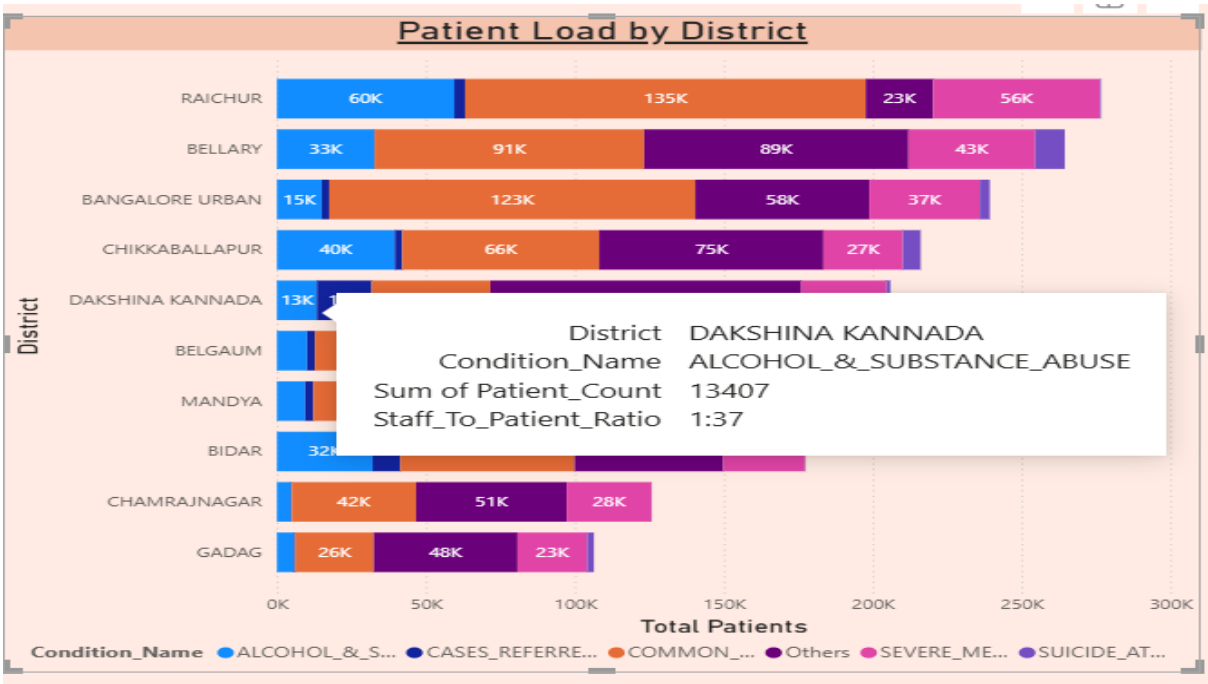
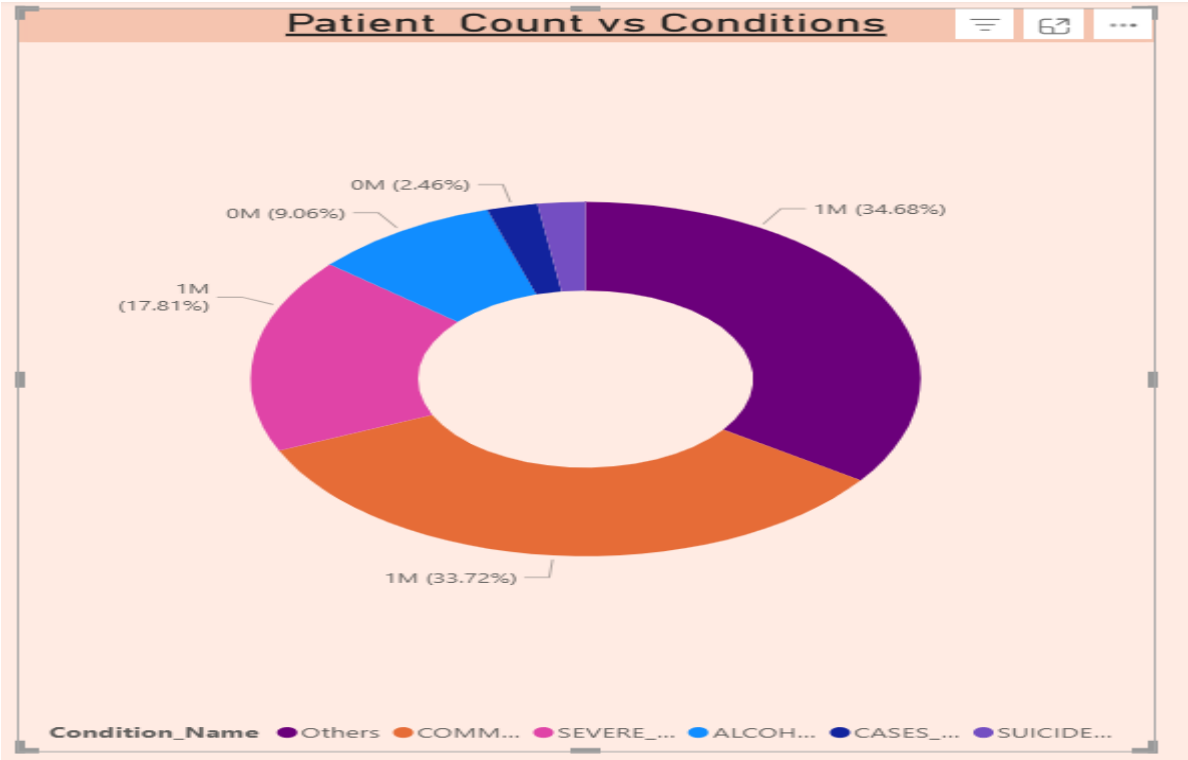
8. Insights & Conclusions

- **Key Findings:**
  - **Massive Growth in Cases**: The total number of psychiatric patients in Karnataka skyrocketed from roughly 3.6 lakh in 2018 to over 11.2 lakh by 2022.
  - **Literacy Paradox**: Found a very weak negative correlation (-0.15) between women's literacy and patient counts. This means that a district having more educated women does not automatically mean it has fewer (or more) mental health patients.
  - **Alcohol Misalignment**: There is almost no correlation (-0.18) between the percentage of the population that drinks and the number of people seeking help for alcohol abuse. Some districts with high drinking rates have very few reported patients.
  - **Top Districts**: Districts like Raichur, Bellary, and Chikkaballapur consistently show the highest patient volumes, regardless of their literacy or alcohol consumption rankings.
- **Analysis Insights:**
- **Total number of Mental Health Incidents recorded across all years**: 34,49,098.
- **Top 10 Districts with highest number of patients**: Patients load in "Raichur" is almost 274K, with Common Mental Disorders of approximately 135K and Alcohol & Substance Abuse at approximately 60K. "Bellary" follows this by a total of 256K, with Common Mental Disorders of approximately 91K and Other conditions of approximately 89K.

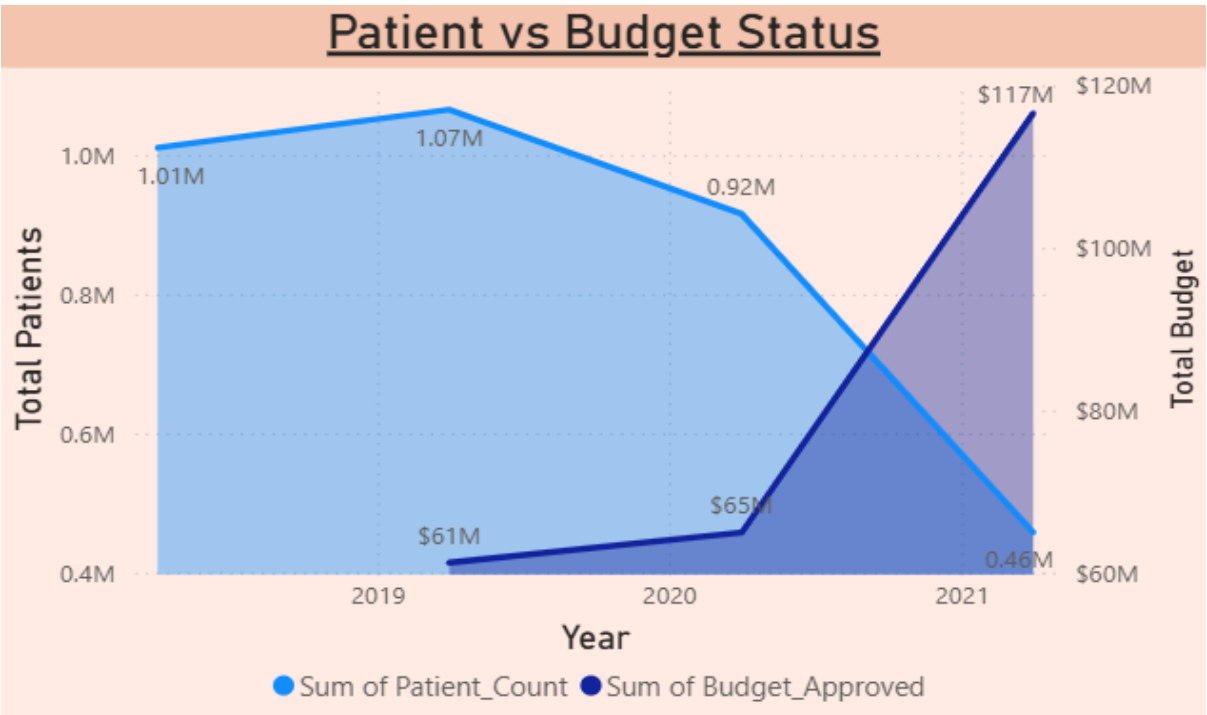


- **Distribution of different Mental Health Conditions (e.g., SMD, CMD, Alcohol Abuse)**: Diagnostic distribution analysis reveals "Other" condition [personality disorder, eating disorder, etc.] is leading by 34.68%, followed by "Common Mental Disorder" condition [depression, anxiety, etc] by 33.72%. This near-equal prevalence suggests a high diversity of mental health needs. Further, the fact that other categories fall below 18% warrants further investigation as to whether this might reflect lower actual prevalence or potential gaps in specialized diagnostic access for the remaining conditions.



- **Budget Change from 2018 to 2022**: Due to data limitation, the annual budget of 2018 was not available; however, subsequent year demonstrate a steady upward trend in budget for mental health issues. Despite the budget growth, the Budget_Per_Patient measure stands at only ₹70.36. When contrasted with the rising patient load and the current staffing ratio of 1:9,608, this figure suggests that the fiscal increase is not keeping pace with the growing demand for services. This indicates a critical need to re-evaluate resource allocation to ensure that the 'per-patient' investment is sufficient to provide quality clinical care and support specialized staffing requirements.

1:9,608		
3449098	242,686,000.00	70.36
Total_Patients	Total_Budget	Budget_Per_Patient



- **Pandemic vs. Pre-Pandemic Incidents View**: By looking at the table below, there was a steady growth in the system, but due to pandemic, the growth dropped by 14% indicating "service gap" where patients could not reach the hospitals. Furthermore, it dropped to 50% indicating that the system has not recovery.

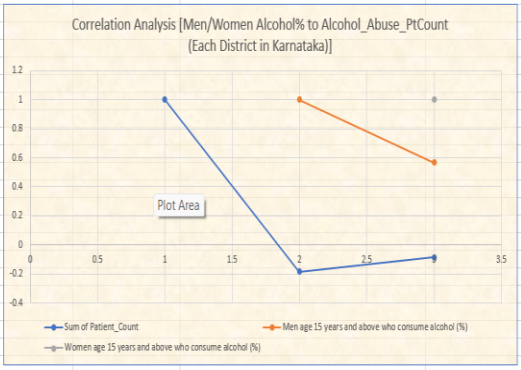
Financial_year	Sum of Patient_Count	YOY % Change
2018-19	1010467	
2019-20	1065175	5.41%
2020-21	915657	-14.04%
2021-22	457799	-50.00%
Grand Total	**3449098**	

- **High Alcohol Consumption vs Alcohol & Substance Abuse Cases**: Men's Alcohol Consumption vs. Patient Count: -0.18 - weak negative correlation; Women's Alcohol Consumption vs. Patient Count: -0.08 - no connection; Men vs. Women Consumption: 0.56 - moderate positive correlation. One might generally expect more people drinking, more patient records; however, the data shows contradiction: Hassan has the highest percentage of men drinking (26.6%), but only 10,186 patients. Raichur has a much lower drinking percentage (15.8%), yet it has the highest patient count by

- far (59,517). This makes the analysis void because there may be external confounding variables [*Population Size, Access Restriction, etc*] in play.

District	Sum of Patient_Coun	Men a	Wome
BAGALKOTE	757	14.8	0.6
BANGALORE RURAL	3416	17.8	1.5
BANGALORE URBAN	15054	17.3	1.1
BELGAUM	10128	11.5	0.5
BELLARY	32671	14.3	1
BIDAR	32135	16.6	0.3
BIJAPUR	1435	15.3	0.8
CHAMRAJNAGAR	4684	20	0.9
CHIKKABALLAPUR	39585	11.7	1.2
CHIKMAGALUR	1964	23.8	1.5
CHITRADURGA	4194	21.2	2
DAKSHINA KANNADA	13407	11.6	0.6
DAVANAGERE	5746	14.1	1.7
DHARWAD	1292	15.8	0.5
GADAG	5699	13.7	0.5
GULBARGA	10687	17	0.8
HASSAN	10186	26.6	1
HAVERI	4933	12.1	1
KODAGU	3042	27.9	2.3
KOLAR	7588	12.8	0.3
KOPPAL	1871	13	1.2
MANDYA	9505	19.1	0.7
MYSORE	14815	22.2	1.6
RAICHUR	59517	15.8	0.9
RAMANAGAR	2538	17.9	0.5
SHIMOGA	4288	18.5	1.4
TUMKUR	3035	18.8	0.8
UDUPI	2501	15.6	1
UTTARA KANNADA	1501	12.3	0.3
YADGIR	4164	19.4	1.1

Alcohol_Correlation_Analysis	Sum of Patient_Cou	Men age 15 years and above who consume alcohol (%	Women age 15 years and above who consume alcohol (%
Sum of Patient_Count	1		
Men age 15 years and above who consume alcohol (%)	-0.18200546	1	
Women age 15 years and above who consume alcohol (%	-0.085595778	0.56803012	1

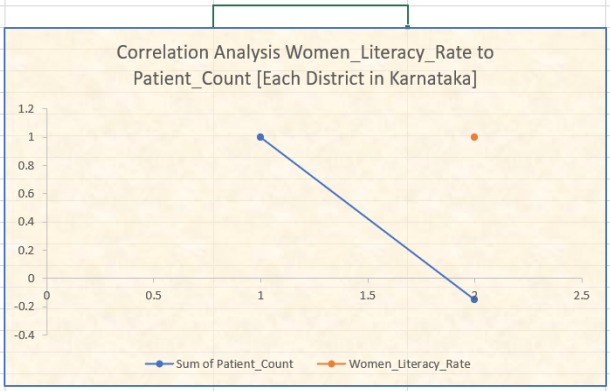


- **Lower Literacy Rates vs Higher Severity of Mental Health Cases**: Negative correlation - With reference to the -0.15 correlation, no meaningful relationship between these literacy rates and severity of mental health cases are noticed. No data pattern is noticed. This is backed up by the following: a) Yadgir has a low literacy rate (48.1) and a relatively low patient count (50,608). b) Raichur has a slightly higher literacy rate (54.3) but a massive patient count (276,688). c) Tumkur has a high literacy rate (81.9) and a low patient count (96,286).

  [Disclaimer: The correlation may be ineffective due to confounding variables like total population to total cases registered, unregistered cases, unavailability of mental healthcare or mental health professionals, etc]

District	Sum of Patient_Count	Women_Literacy_Rate
YADGIR	50608	48.1
RAICHUR	276688	54.3
KOPPAL	97223	59.8
BELLARY	264525	64.4
BIJAPUR	46722	66.6
GULBARGA	100501	68.2
BAGALKOTE	54680	69.7
GADAG	106414	70.7
HAVERI	95570	71.5
CHAMRAJNAG	125851	72.4
BIDAR	177440	73.8
BELGAUM	187236	74.0
CHITRADURGA	102670	75.6
DAVANAGERE	57480	76.0
CHIKKABALLAF	216112	76.3
KOLAR	92752	78.3
MANDYA	183320	78.3
MYSORE	81869	78.9
SHIMOGA	82898	79.8
DHARWAD	78321	81.8
TUMKUR	96286	81.9
HASSAN	91959	82.0
RAMANAGAR	54209	82.7

Lit_Correlation_Analysis	Sum of Patient_Count	Women_Literacy_Rate
Sum of Patient_Count	1	
Women_Literacy_Rate	-0.151537557	1

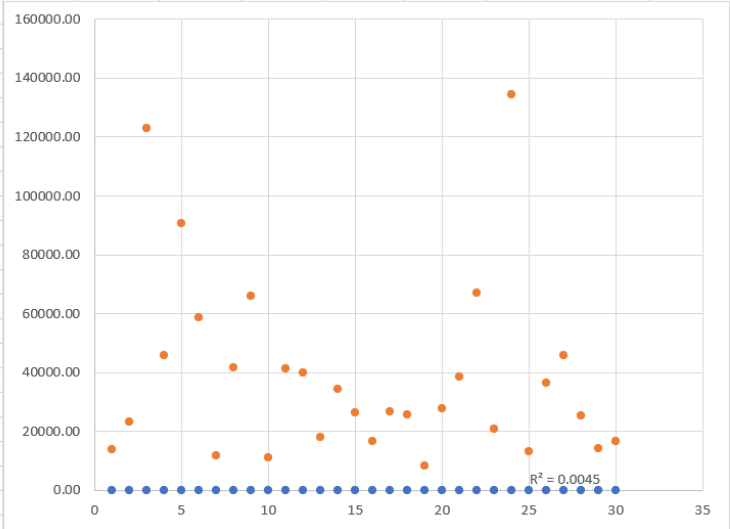


- **Higher BP vs Higher Cases of CMD**: Value of -0.0714 clearly indicates a negative correlation between Common Mental Disorders and High Blood Pressure.

[*Disclaimer: The correlation may be ineffective due to confounding variables like population size, diet, other health issues, exercise, etc*]

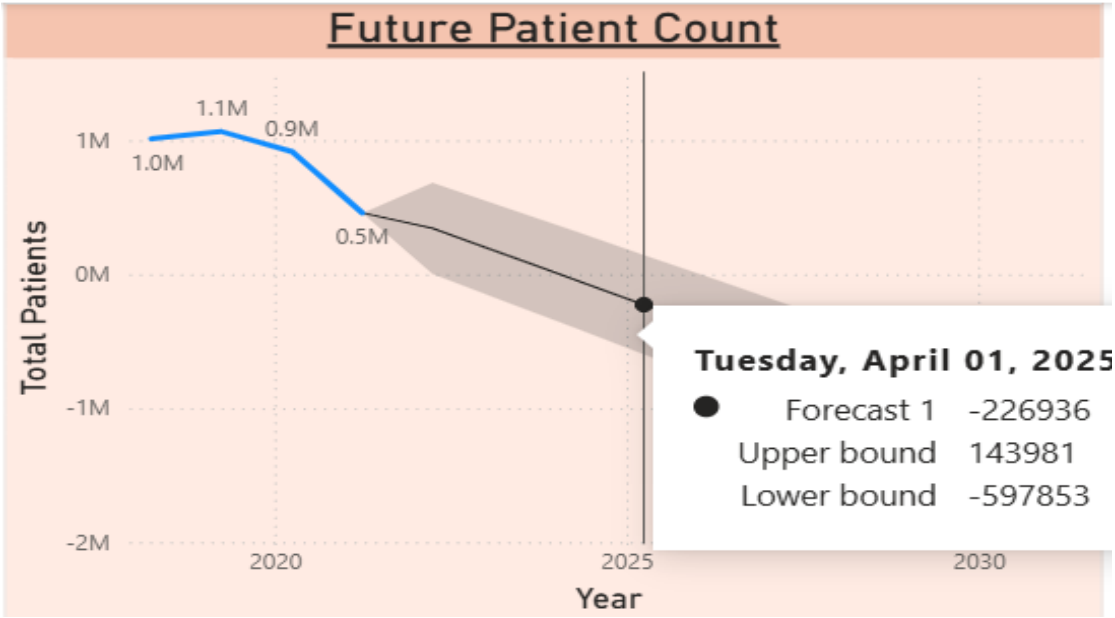
Row Labels	Avg_BP (%_	Patient_C
BAGALKOTE	20.05	13819
BANGALORE R	28.75	23237
BANGALORE U	32.10	122962
BELGAUM	20.80	45866
BELLARY	21.05	90511
BIDAR	22.65	58569
BIJAPUR	20.60	11744
CHAMRAJNAG	29.35	41760
CHIKKABALLAF	28.30	66127
CHIKMAGALUF	28.95	11114
CHITRADURGA	30.15	41448
DAKSHINA KAN	30.00	39902
DAVANAGERE	27.15	18034
DHARWAD		34440
GADAG	23.00	26298
GULBARGA	20.90	16462
HASSAN	28.80	26760
HAVERI	22.90	25575
KODAGU	32.40	8381
KOLAR	26.10	27728
KOPPAL	20.85	38349
MANDYA	29.95	66995
MYSORE	26.45	20775
RAICHUR	17.60	134573
RAMANAGAR	29.05	13274

-0.0714



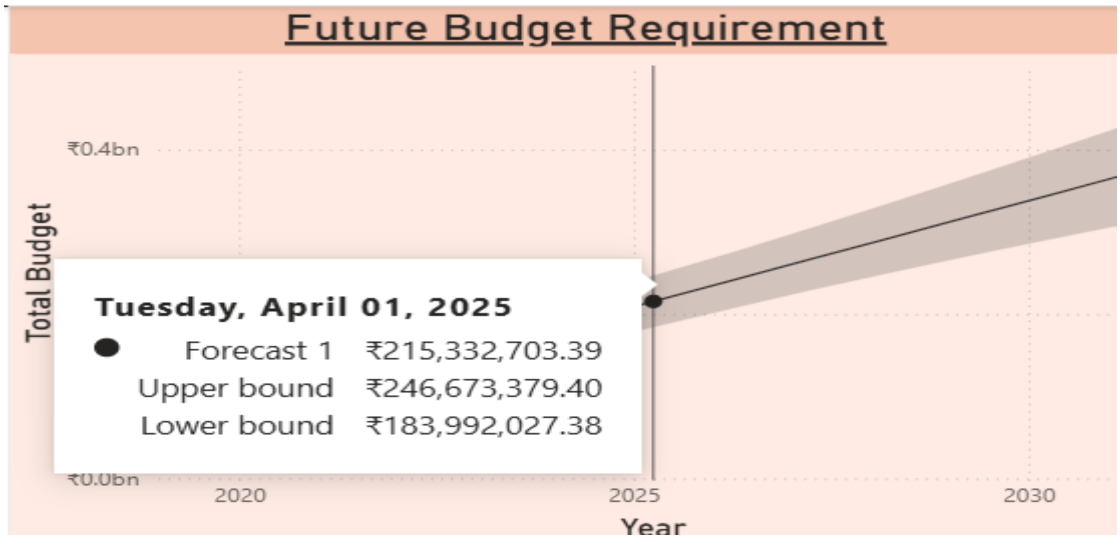
- **Patient-to-Estimated Staff Ratio**: Raichur, Bellary, and Bangalore Urban, with the high incidents reported, shows significant strain on estimated staff capacity.

- **Future Incidents Forecast**: Forecast reveals a significant downward trend in recorded incidents. While the Forecast Value predicts a continued drop, the Upper Bound reminds us that we must remain prepared to handle at least 1.4 Lakh patients. The wide gap between the Upper and Lower bounds suggests high uncertainty, likely due to the massive fluctuations in patients reporting during and after the pandemic.

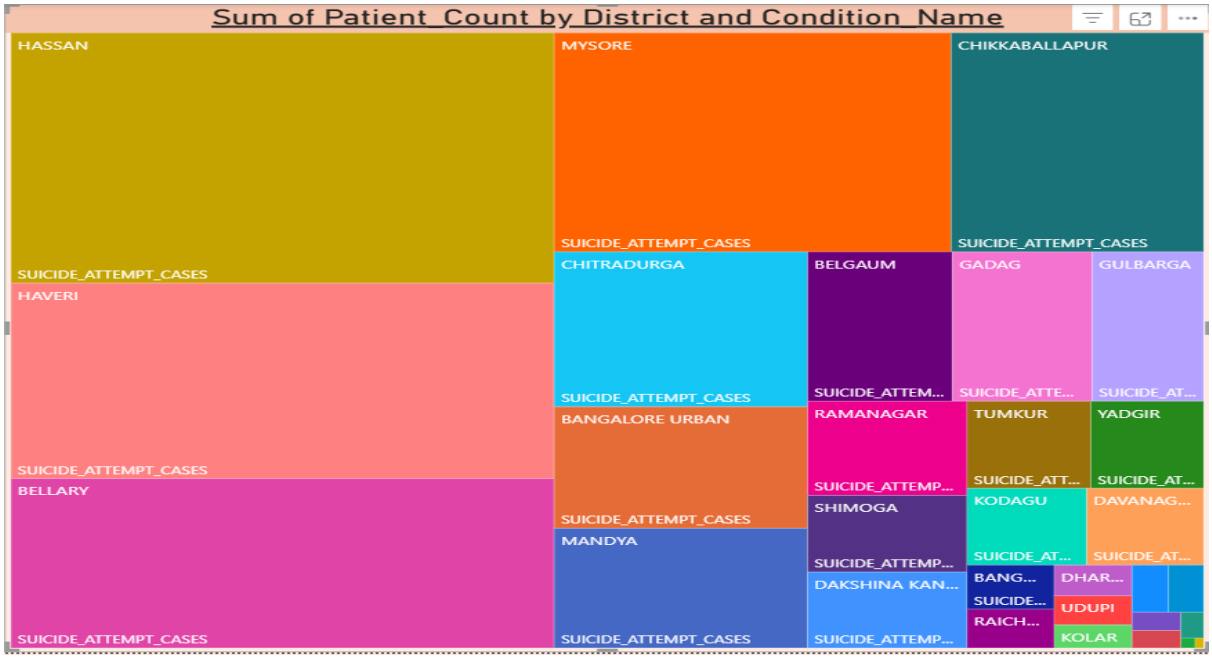


- **District exceeding Estimated Capacity based on Incident Growth**: Our prediction suggests that Raichur, Bellary, and Bangalore Urban are the 'Strategic Priorities' for following years. Without a budget reallocation to increase the ₹70.36 per patient spend, these districts will likely exceed their healthcare capacity.

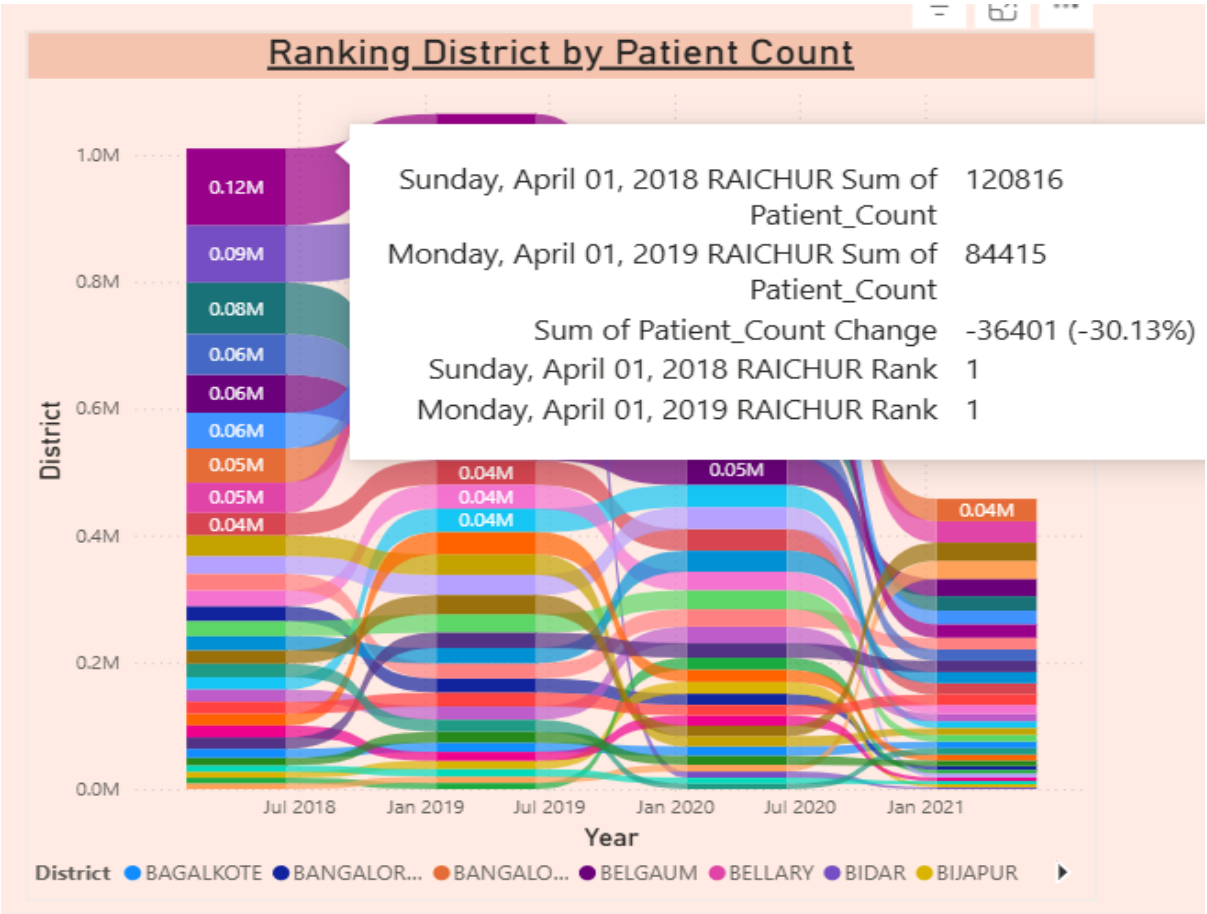
- **Future Budget Allocation**: If the budget continues to grow at its current rate, approximately ₹21,53,32,704 for the fiscal year 2024-2025; it still raises concern due to existing patient load and low spent of ₹70.36 per patient.



- **Highest Suicide Cases vs "Fair Share" Budget**: Analyzing the data, Hassan, Havery, and Bellary likely have the highest suicide cases registered. Hassan and Havery have less than 100K registered patients when compared to Bellary of approximately 256K patients. Their "Fair Share" budget [1/30] is being spread too thin to provide suicide prevention programs.



- **Deployment of Psychiatric Social Workers based of Patient Load**: Comparing the ribbon and bar charts, Bellary, Bangalore Urban and Raichur stands firm on their patients load, this indicates increasing psychiatric social workers in these districts.



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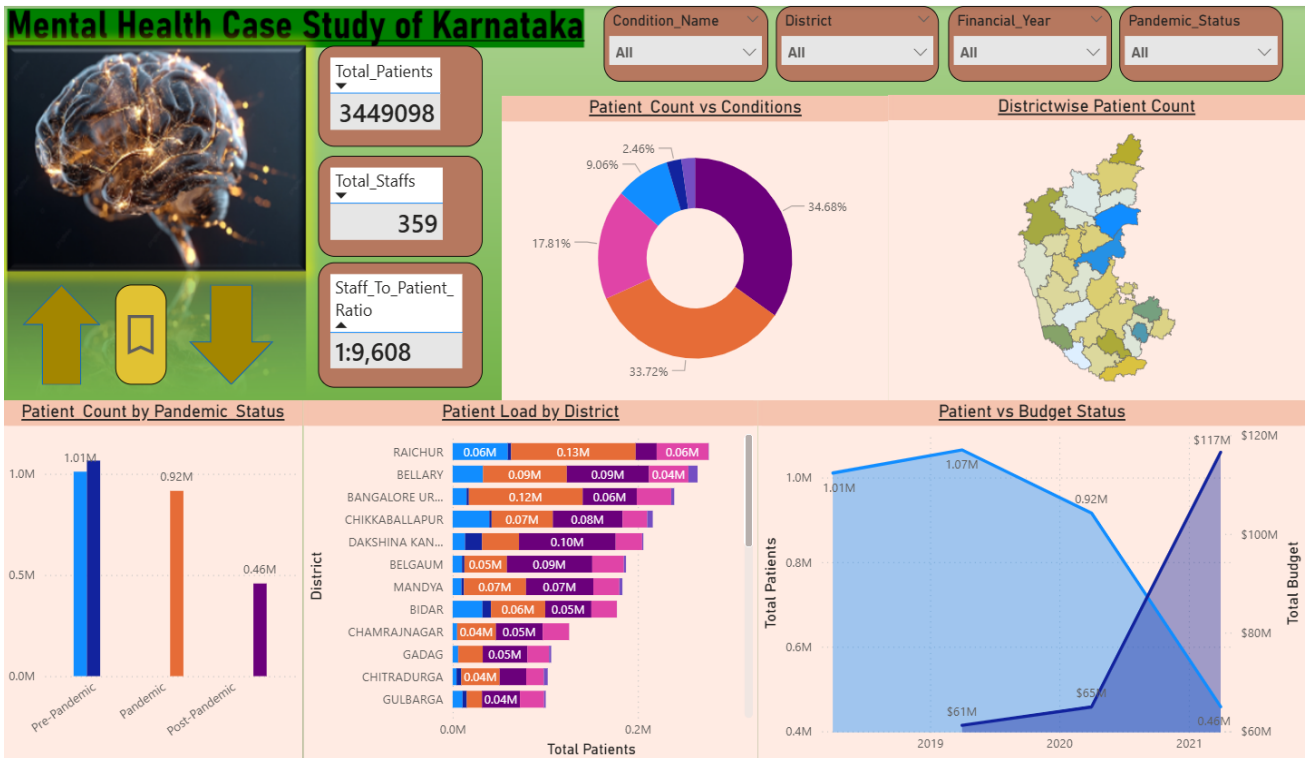
9. Consolidated Report/Dashboard



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10. Conclusions

The most important takeaway is that social statistics (like literacy or drinking %) are not currently predicting hospital visits. This suggests a "Gap in Care". High patient counts in certain areas likely reflect better hospital infrastructure or better reporting systems, rather than just "more sickness". Conversely, districts with high alcohol use but low patient counts may suffer from a lack of accessible mental health facilities or high social stigma that prevents people from seeking help.

The following strategies are recommended to provide better mental health services:

a) Normalize Data Population - to get a true sense as to which districts are in crisis, the State must move from “Total Sums” and move towards analysing “Patient per 100,000 residents.”
b) Invest in Low-Reported Districts - Districts with low literacy and low patient counts should be targeted as they may have hidden patient populations who don’t have access to mental health education or transportation.
c) Targeted Alcohol Intervention - Since alcohol consumption rates between Men and Women are somewhat linked, Alcohol Awareness Programs should be implemented to families rather than targeting individuals.
d) Standardized Reports - to improve further data collection, there need to be a standardized and uniform data collection and reporting through the State.

The statistics observed from analyzing the data, significant gap between social status and healthcare access is observed. By focusing on the infrastructure that allows patients to be counted, Karnataka can move toward a more equitable mental health strategy.